

The Lean formalization of Gwynne–Miller’s LQG metric theorems: what could not be repaired as printed

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Headline. No irreparable flaw was found in the proof strategy. Theorems 1.1 and 1.2 of Gwynne–Miller, *Existence and uniqueness of the LQG metric for $\gamma \in (0, 2)$* (arXiv:1905.00383, “GM”), are proved in Lean 4 with no hypotheses and only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`); this was kernel-checked twice. Every cited result that GM’s proof uses (from CONF, DG, DZZ, DDDF, DFGPS, LM, MQ, DGo) is proved in Lean, some in corrected form. Nothing below invalidates GM’s theorems or their argument. Listed are the printed statements that are false and the steps that cannot be completed as written, with what is proved instead.

1. Statements that are false as printed

- **GM Lemma 4.18** (l. 2284–2318, (4.18)). Off by one: $K = m = 1$, $\alpha = 0.99$, $\delta = 0.98$ and independent events of probability 0.98 give $0.9604 > e^{-1.92}$ (the sum has $K + 1$ terms, the threshold $K - m$ does not). Proved with the thresholds counted over $K + 1$ indices; the only use, Lemma 4.21, needs only e^{-cm} decay.
- **GM Lemma 5.6**, the tube $V_r(z)$ (l. 2945; likewise Lemma 5.8). Built from squares of $\mathcal{S}_{\varepsilon r}(B_{2r}(z))$, it misses $z - 2r$ when $z - 2r$ lies on a grid line, although $z \pm 2r \in V_r(z)$ is claimed. Proved with the squares of $\mathcal{S}_{\varepsilon r}(\overline{B_{2r}(z)})$, as GM’s construction (l. 2982–2989) allows.
- **GM Lemma 3.9** (l. 1397–1404): for every open U , w.h.p. every $z \in \setminus U$ has r and a grid point $w \in \setminus U$ with $z \in B_{r/2}(w)$ and $E_r(w)$. False for unbounded U and for z near $\partial(\setminus U)$. Proved for bounded U with w within $\varepsilon^{1+\nu}\setminus/100$ of z , which is what the proof gives and Section 3.7 uses.
- **GM Section 3.3** (l. 1263): “the event described in (A) is contained in $\overline{G}_r(C', 1 - \alpha)$ ”. The points $v \in \partial B_r(0)$ of (A) are not in the open ball over which $\overline{G}_r$ is defined. Repaired by applying Proposition 3.4 with $C_1 = (C' + C_*)/2$ and moving v radially inward by continuity.
- **Also false as printed in GM**, each with a routine repair: the count $\#\mathcal{K}_\varepsilon^\setminus = \lfloor \nu \log_8 \varepsilon^{-1} \rfloor$ of §3.2 (l. 1303; fails at $\varepsilon = 1/8$, $\nu = 1/2$); Remark 4.10 (l. 1975; the two events use different random normalizations); and the statement of Lemma 4.14 (l. 2101), whose compact connected $Y \subset \mathbb{C} \setminus \mathcal{K}$ cannot exist (GM’s own construction is an open arc).
- **CONF Lemma 2.14** (C:832–838). The bound $(1 - AC^{-2} \text{area}(U))n$ is false for small n (the proof discards up to 8 arcs, C:855; a disc with a thin fjord, $n = 1$), and as printed the conclusion is trivially satisfiable, too weak for Lemma 2.15. Proved with an additive loss N_0 and the disconnection within $Cn^{-1/2}$ of ∂U ; Lemma 3.10 needs only $n_k \geq N_0$.
- **CONF Lemma 3.3**, the event G^U of (3.9) (C:1205, 1208, 1219). For “pocket” configurations (two columns of removed squares with a thin opening) U is connected but $U_{\delta r/4}$ is not, so the internal distances in (3.9) are infinite, $G^U = \emptyset$, and (3.12) and the path claim at C:1208 fail. Replaced by an event on the grid components of the non-removed squares, which gives (3.21).
- **DG Proposition 3.16**, lower bound (DG:1436). Fails with probability one: for z, w at distance 2ε on either side of a side shared by two grid squares, $D^\delta(z, w) \rightarrow 0$ as $\varepsilon \rightarrow 0$ while $\widehat{D}^\delta(z, w) - \delta e^{\xi \widehat{h}_\delta(v_{S_z})} \geq \delta e^{\xi \widehat{h}_\delta(v_{S'})} > 0$ (kernel-checked: `not_dgProp3_16Printed`). The statement with error term $\delta^{1-\zeta} e^{\xi \max \widehat{h}_\delta}$ (DG’s own count, DG:1505–1508) is proved from DG Lemma 3.7 and is all Proposition 3.17 uses.
- **DZZ (eq-cell-LQG-compare)** (l. 1181). Over all $S \in \mathcal{S}_B$ it fails at $\partial \mathbb{V}$, where squares outside $\mathbb{V}$ have mass 0. Proved for $S \subseteq \mathbb{V}$, all the proof uses.
- **DFGPS Proposition 4.1**, Steps 1–2 (T:2464–2475). “Each path from u to v in $B_\varepsilon(\setminus L)$ must

enter $B_{2\varepsilon\setminus}(z_k^\varepsilon)$ for each $k \in [k'_1 + 2, k'_2 - 2]$ ” is false when L is a whole circle: a path may go round either way. The block of centres is chosen per path; the union bound over pairs (k_1, k_2) is unchanged.

2. Printed steps that cannot be completed as written

- **GM Section 2.4(e)** (geodesic confinement) is used without proof and does not follow from Axiom V (proved from DFGPS Thm 1.5, LM L3.1). **GM Lemma 4.14**, proof (l. 2482): “every unbounded connected subset of $\mathbb{C} \setminus \mathcal{K}$ whose closure contains y meets X ” fails for $y \in \overline{X}$.
- **CONF Lemma 2.10** (C:722–731). The white-noise approximation needs $h_{t,\infty}$, extended by 0, to be continuous, which can fail for irregular U . Proved for the domains CONF use (an annulus minus finitely many grid squares), via the heat-kernel representation of the Dirichlet Green function there.
- **CONF Theorem 3.9** (C:1506–1509 with Lemma 3.6 Step 3, C:1431). The filled ball $\mathcal{B}_\tau^\bullet$ of a stopping time is treated as determined by h modulo additive constants. It need not be: $\tau := \tau_\setminus$ on $\{B_{\setminus/10} \subseteq \mathcal{B}_{\tau_\setminus}^\bullet, h(\psi) > 0\}$ ($\int \psi = 1$), $\tau_{2\setminus}$ otherwise, is a stopping time whose ball is not local modulo constants. Added as a hypothesis, verified for every stopping time CONF and GM use. At C:1559–1561, “ $\mathcal{I}_k$ is determined by $(\mathcal{B}_{s_k}^\bullet, h|)$ ” has no measurability argument (the arcs use the leftmost geodesic, unique only a.s., so the sets are a priori Σ_2^1); repaired by an a.s. transfer to GM’s analytic arc relation (GM l. 1705–1708).
- **DZZ Corollary 3.3** (l. 859–862). The rate $e^{-(\log \delta^{-1})^{0.9}}$ does not follow from Proposition 3.2 ($|\log D - \log D'| \leq (\log \delta^{-1})^{0.9}$ w.h.p.); proved with rate $(\log \delta^{-1})^{-0.1}$; every consumer only needs the difference to vanish.
- **DZZ Lemmas 5.4 and 6.1** (l. 2559–2567, 2605). Segments of length $\delta^{2\iota}$ with crossing cost $\delta^{-\chi+\iota}$ close the gluing only when $\chi \geq \frac{1}{2}$; length $\delta^{K\iota}$ with $K\chi \geq 2$ is used, and the leg at l. 2563 is confined to $\tilde{\mathbb{V}}_{u,v}$. **Remark 5.2** asserts the walled Propositions 3.2/3.17 with no range of pairs; read literally they fail near the wall. Proved for pairs at distance $\geq \xi$ from the complement, the only ones DZZ use.
- **DZZ Lemma 5.3, part 1** (l. 2490–2492, 2474, 2533). “Corollary 3.9 and Proposition 3.17 give $O(K^{-4})$ ” fails at $\tilde{\delta} = 2^{-l}$ with l small (they hold only with probability $1 - \tilde{\delta}^c$, resp. $1 - e^{-(\log \tilde{\delta}^{-1})^{0.7}}$); replaced by concentration at $\tilde{\delta}e^{-L^{0.95}}$ and a two-regime comparison of means. Close pairs are not covered ($a \asymp s_i^{-1}K$ presumes $|z - z'| \asymp s_i/K$); they are declared far.
- **DFGPS Proposition 4.3** (T:2699–2741). Excursions start at $\tau_0 = s$ and only geodesic times $\geq s$ are bounded; earlier times need a point of ∂B_k at distance $\geq s$ from 0, which fails when B_k surrounds a hole of $\mathcal{B}_s$. Proved for filled balls (backward excursions work), the only case GM use.
- **DFGPS Lemmas 2.14 and 2.17**. The limit $\mathfrak{c}_r = \lim r\mathfrak{a}_{\varepsilon/r}/\mathfrak{a}_\varepsilon$ (T:1073–1098) needs joint convergence with h and a unique median of the internal crossing, neither argued; a positive cluster point with the Λ -bounds is proved, all Lemma 2.13 uses. In Lemma 2.17 Step 4 (T:1276–1277), conditioning on $h|_V$ does not pass to $h|_{\overline{V}}$ from (eqn-limit-metric-ind) (two-bit counterexample); $\sigma(h|_{\overline{V}})$ is carried from the LFPP stage.
- **LM Theorem 1.7**, Step 3 (l. 1075–1084). The bound $\text{len}(P_\varepsilon \cap S; D) \leq \sup_S D + \varepsilon^2$ fails because the near-geodesic of $D(\cdot, \cdot; V)$ must stay in V . Proved with $U = \mathbb{C}$, $V = B_n$, Steps 1–2 unchanged and Step 3 replaced by a conditional-variance bound that vanishes as $n \rightarrow \infty$.

Each repair is local; GM’s Theorems 1.1 and 1.2 are proved as stated.